



Module 9

Resistance-Training Theory

For many endurance athletes, strength training is not done at all or only during the off-season. Strength training is extremely beneficial from a performance standpoint, and those who do not actively incorporate it into their training program are at a significant disadvantage to those who do. Runners who perform strength training only in the off-season will lose the benefits gained as the season progresses. In this case, the saying *use it or lose it* is very applicable. People would not expect that if they did cardiovascular training for only four months that the results gained would last for the other eight months of the year. Despite this, many endurance athletes take this approach to strength training.

Strength training can be interpreted in many different ways. For example, in theory going for a run constitutes strength training, as there is tension on the muscles. Others consider strength training a workout only if they are in a gym lifting weights. For the purpose of this certification, resistance training relates to exercises that may or may not utilize weights or strength-training equipment. Moreover, resistance training equates to any type of resistance on muscles for the purpose of performance enhancement via strength and/or power gains.

The terms *resistance training* and *strength training* are considered synonymous and thus are used interchangeably.

❖ OBJECTIVES

Upon completion of this module, you should have an understanding of the following areas:

- USECA's standpoint on resistance training
- Why resistance training?
- Myths of resistance training
- Physiology of resistance training
- Resistance-training protocol
 - Areas of focus
- Categories of resistance training
- Programming
- Integration versus nonintegration of resistance training

❖ WHY RESISTANCE TRAINING?

Just as it is important to understand why runners resistance train, it is equally important to understand why many runners do not do it. In this section, both of these areas will be discussed.

UESCA's Standpoint on Resistance Training

- Resistance training is a critical aspect to a running program, specifically in regard to improving running economy.
- The majority of endurance athletes have pronounced muscle imbalances, weak core (inner unit) and upper-body musculature. Therefore, resistance training is important for all runners.
- Resistance training is just as important as cardiovascular training.
- Resistance training can help to maintain lean muscle mass.

Improve Running Economy

A 2008 study of distance runners by Storen et al. found that heavy weight training over an eight-week period improved the economy and time to exhaustion of the runners (no change in VO_2 max or body weight) (654). This study

correlates with the results of a study by Paavolainen et al. (282) that found explosive strength training improved 5K running times by improving running economy.

A study by Mikkola et al. prescribed three groups of recreational runners varied strength-training programs, all while concurrently running (treadmill-based). The three strength-training groups were:

1. Heavy resistance
2. Explosive resistance
3. Light to medium resistance

All of the runners improved their running performance. However, only the heavy and explosive resistance-training groups improved their neuromuscular fitness levels (655).

Do It Right or Not at All

While correct form is important during all aspects of training, it is absolutely critical for resistance training.

It is your responsibility as a coach to be cognizant of what constitutes correct and incorrect form and accept nothing less than correct form. Incorrect form greatly increases the chance for injury. Correct versus incorrect form is often measured in millimeters or degrees. For example, picking up a weight with a flat lumbar spine and stable core musculature is a safe movement. Add a few degrees of flexion of the lumbar spine and it can become extremely dangerous because of shearing forces on the vertebrae. For exercises that involve multiple joints and/or planes of movement, the task of ensuring correct form becomes significantly more difficult than single-joint movements.

If you notice your clients using incorrect form, stop them immediately and correct it. **There is no such thing as “slightly incorrect form.”** It is either correct or incorrect. Allowing your clients to proceed with bad form not only risks injury and diminishes the effectiveness of the exercise, but it essentially tells clients that the incorrect form is correct since they were not told otherwise. Do not allow and reinforce incorrect movement patterns.

Time Constraints

Many runners do not strength train because of a lack of time. While training for distance running can be a time-consuming endeavor, most runners look at strength training incorrectly from a time perspective. First, strength training should not be viewed as an *if I have a little extra time I'll do it* sort of thing. Second, strength training can be done anywhere at any time. A runner does not need a gym to do strength training. For example, clients could do a wall sit exercise while waiting for their eggs to cook.

Strength training can be done throughout the day, and therefore it is not necessary to lump it into one set time frame. The key to an effective program is being flexible and appreciating the “whole picture” when designing a program. If this means spreading 10 strength-training exercises throughout the day, so be it – as long as it does not negatively affect other aspects of the training program. Integrating strength training into a running program is discussed later in this section under *Concurrent Training*.

Movement Planes

Body movements of runners primarily occur in the sagittal plane and transverse plane (hips, torso). If runners have not trained for movements outside the sagittal plane, when they are forced to move in other planes, the strain on the muscles, ligaments, and tendons often results in soreness at best and injury at worst.

While the body does not move much in the frontal plane while running, there is a need to stress the muscles in this plane as many stabilizer muscles are strengthened this way. These stabilizer muscles help to support the primary mover muscles during sagittal plane movements. Therefore, while frontal plane movements may not mimic sport-specific movements, the strengthening of stabilizer muscles will help to enhance overall running performance.

When running, the body expends energy not only to move forward, but also to stay upright and balanced and absorb ground reaction forces. In regard to the percent of one's body weight, 250 percent is absorbed vertically upon each foot strike, 50 percent is used to stabilize in the sagittal plane (resist falling forward/backward), and approximately 20 percent is used to stabilize in the frontal plane (resist falling over sideways) (669). This requires runners to be able to control their body, in multiple planes of movement.

Cardiovascular Fitness Precedes Musculoskeletal Readiness

This concept is mentioned in other sections of this certification and means that one's cardiovascular system adapts to training faster than the musculoskeletal system. This is likely a large reason for the high injury rate among runners. For example, just because someone has the cardiovascular fitness to run 20 miles does not necessarily mean the person's muscles, bones, and connective tissue are ready for this sustained impact.

By performing specific resistance-training exercises, a runner can speed up the musculoskeletal adaptation to reduce the chance of injury. While it is unlikely that even with strength training a runner can adapt at the same speed in both cardiovascular and musculoskeletal areas, it will likely reduce the adaptation gap between the two.

MYTHS SURROUNDING RESISTANCE TRAINING

For the purpose of this certification, it would be advantageous not to think of Arnold Schwarzenegger in the movie *Pumping Iron* when considering resistance training. While for some runners putting on muscle might be the goal, it is certainly not the goal to become a competitive bodybuilder or power lifter. The goal of resistance training is to go faster and get stronger, not necessarily bigger – or at least not substantially so.

One pervasive theme within the endurance sports community is that resistance training will add muscle bulk and therefore should be avoided at all costs. This common misconception is often rationalized by using two erroneous validations:

1. Muscle bulk will slow me down
2. Power-to-weight ratio will decrease

Because of the two reasons listed above and for many other invalid reasons, most distance runners never maximize their potential in regard to strength and power. The focus is often so high on the cardiovascular facet that all other aspects of fitness go out the window. The question to these runners is that while they may be aerobic monsters, how much faster could they be if they had a stronger core or fewer muscle imbalances? The reality is that by not doing

strength training, any gains in cardiovascular fitness are marginalized because the athlete is missing out on a major fitness component.

In regard to misconception #1 (strength training will add muscle bulk), a 2015 study by Beattie et al. found that over a 40-week period, distance runners who participated in a strength-training program were able to improve their leg strength while not adding any “unwanted” muscle mass. It is proposed that this adaptation is largely due to improved neuromuscular function (692).

Because strength training is either negated completely or performed in minute quantities by most runners, it will become the secret weapon to your client's success. Resistance training done with the right selection of exercises and with proper form and progression can greatly improve the performance of a runner.

Anyone who has tried to put on a lot of muscle while maintaining an intense cardiovascular endurance-based program will tell you it is next to impossible. The myth is that if resistance training is performed, all of a sudden you will look like a bodybuilder. This could not be further from the truth. To put on substantial muscle size (*hypertrophy*), you would have to eat a huge sum of calories each day and do specific weight-training exercises in set cycles that would not resemble an endurance-based strength-training program. Additionally, you would have to do a minimal amount of cardiovascular exercise, as this inhibits muscle hypertrophy through the burning of calories that could be going to building muscle. As you can see, this is hardly a runner's program.

The *power-to-weight* argument is synonymous with putting on muscle bulk. In actuality, by performing strength training specific to endurance sports training, one's power-to-weight ratio will increase. Again, the assumption by those who feel strength training will decrease their power-to-weight ratio is they will put on substantial amounts of muscle.

Genetics do come into play to some degree when discussing muscle hypertrophy. For example, if a client has a large proportion of fast-twitch muscle fibers, the person will likely increase muscle mass to a greater extent than someone with predominantly slow-twitch fibers. However, assuming the individual is eating a normal amount of calories, strength training for muscular endurance and performing a substantial amount of cardiovascular exercise, the actual hypertrophy would be minimal.

PHYSIOLOGY OF RESISTANCE TRAINING

Overload Principle

Benefits reaped from strength training are the result of the *overload principle*. The overload principle states that a greater-than-normal stress on the body needs to be present for a training adaptation to occur (108). The overload principle was developed by Thomas Delorme, M.D., who used heavy-resistance training exercises on soldiers who had just returned from WWII. Rapid rehabilitation of soldiers was necessary as there was a shortage of hospital beds at that time. In 1951, Dr. Delorme and Arthur Watkins published their findings in a book called *Progressive Resistance Exercises* (113).

While the overload principle originated from a resistance-training perspective, all training adaptations are based on this principle. In regard to strength training, the overload principle can relate to different physiological adaptations, such as muscular endurance or power.

Neuroendocrine Response

An area rarely discussed in regard to strength training is the neuroendocrine response. This response refers to the interactions between the nervous and endocrine systems that occur when performing specific types of strength-training movements. The greatest neuroendocrine response occurs when performing exercises that recruit large muscle masses and are done at a relative high volume with moderate to high resistance (109, 189). This response increases testosterone and hormones (i.e., growth hormone) that stimulate muscle growth/strength.

Testosterone is a hormone within the body responsible for muscle growth (20). Testosterone is a steroid hormone from the androgen group that is responsible for protein synthesis (12). Therefore, testosterone is responsible for an increase in muscle size/strength, hair growth, and bone density. In males, testosterone is secreted by the testes, and it is secreted by the ovaries in females. Additionally, a very small amount of testosterone is secreted by the adrenal glands. Adult males produce approximately seven to eight times more testosterone than females (13).

While the strength program of a runner won't contain high neuroendocrine response exercises exclusively, it is of crucial importance to include them in a program. Exercises that have a high degree of neuroendocrine response are those that incorporate multiple large-muscle groups such as the squat, dead lift, and other Olympic-style lift exercises (20).

The neuroendocrine response also influences the order in which exercises should be completed. Exercises that have a high neuroendocrine response increase, for a short time, the amount of testosterone circulating in the blood. Therefore, performing exercises that do not elicit a high neuroendocrine response (e.g., shoulder raise) should typically be done after exercises that do have a large response. These smaller muscles (low neuroendocrine response) will be able to reap some of the benefits of the response because of the increase in testosterone in the bloodstream.

Muscle Hierarchy

Muscles are often categorized as *primary* and *secondary* movers:

- **Primary Mover:** is a muscle or muscle group that is primarily responsible for producing a movement
- **Secondary Mover:** (also termed “stabilizer”) is a muscle or muscle group that contributes indirectly or secondarily to produce movement. A secondary mover supports/assists the primary mover. For example, during a push-up, the pectoralis major is the primary mover and the triceps are the secondary movers.

RESISTANCE-TRAINING PROTOCOL

Resistance training is integrated into a training program for many different reasons. Some of these include:

1. **Corrective:** Target specific muscle(s) for the purpose of correcting postural issues and muscle imbalances.
2. **Sport Replication:** Exercises that mimic running movements.

3. Increase Power: Exercises are typically called *plyometric exercises*, meaning explosive-type movements.

4. Increase Strength: While all exercises increase muscle strength, not all strength exercises fall into the just-noted categories. For example, you might prescribe a *wall sit* to a client for the purpose of increasing quadriceps strength but not necessarily for the purpose of correcting a muscle imbalance, replicating a sport movement, or to increase power.

The *inner unit core musculature* must be strong in order for all other exercises to be safe and effective. Resistance training can functionally strengthen the LPHC.

An exercise can fall into more than just one category. For example, depending on how and why it is performed, a squat can fall into both the *corrective* and *strength* areas.

Assuming your client has no injuries or physical limitations, for exercises that can be performed seated or standing, it is preferable to do them standing. This puts a greater emphasis on core engagement and the kinetic chain.

The exercises you select for your clients should be based on their specific needs.

Determining Workout Structure

Most resistance-training programs for sports, as well as general fitness, typically assign a range of 10 to 15 repetitions with a two- or three-set range. Why?

Repetition (Rep): One full exercise movement. Performing one repetition at maximum intensity is referred to as *one repetition maximum*, or *1-RM*.

Set: A group of predefined repetitions.

Strength-Endurance Continuum

The genesis of the 10- to 15-rep, two- to three-set exercise prescription is the *strength-endurance continuum*. This continuum states that strength and endurance are inversely related (179). In regard to resistance training, strength increases are associated with high loads and low repetitions whereas endurance

increases are associated with low loads and high repetitions. A repetition range of 10–15 repetitions seeks to capitalize on the benefits of both strength and endurance, as the repetition range falls somewhere on the spectrum between what is typically considered primarily strength and primarily endurance.

In a 2002 study by Campos et al. (180), 32 untrained men participated in an eight-week resistance training study to establish the validity of the strength-endurance continuum. The subjects were divided into four groups: High Repetitions (20-28 reps), Intermediate Repetitions (9-11 reps), Low Repetitions (3-5 reps), and the control group (no exercise).

Subjects were assessed pre/post in the areas of strength (1-RM), muscle endurance (max reps performed at 60% of 1-RM), and cardiovascular (VO_2 max, time to exhaustion, pulmonary ventilation, maximum aerobic power).

Not surprisingly, the low-rep group saw the biggest strength gains and the high-rep group realized the highest gains in muscle endurance as well as significant gains in max aerobic power and time to exhaustion. Interestingly, the low- and intermediate-rep groups did not realize any gain in the latter two areas.

Based on this information, repetition range is not something that should be assigned randomly or doled out the same to all clients. The repetition range assigned to an exercise must be purpose-driven and with a specific goal in mind. If you are looking for large strength gains, low to medium repetitions (three to 11-rep range) are advised. For muscular endurance, a higher-repetition range (20–28 reps) is likely the correct route.

Given the rep ranges assigned to the different groups in the study, it could be theorized that the 10- to 15-rep range so commonly assigned to clients falls in between true strength and endurance benefits and thus the benefits are minimized because of a lack of specificity. Further research is needed to determine if this hypothesis is correct.

However, for clients new to strength training, the repetition range should be midrange with a relatively low weight. Specificity of repetition range should come into play only once the muscle and connective tissue adapt to the strength training.

PURPOSE	WEIGHT	REPETITIONS
Large Strength Gains	High	3 to 10
General Fitness	Low to Medium	10 to 15
Muscular Endurance	Low to Medium	20 to 28

Figure 9.1 Strength/Endurance Continuum Chart

ALTERNATE RESISTANCE TRAINING DURATION PRESCRIPTIONS

Rate of Perceived Exertion (RPE)

Another way to guide your client in regard to intensity level is the *rate of perceived exertion (RPE) scale*. Very easy intensity would rank a “1,” and extremely difficult (almost to muscular failure) would rank a “10.” For example, an RPE of “8” would represent a difficult effort but not to the point of muscular failure. The RPE scale is very useful in determining rep ranges, so long as you know that your client’s interpretation of his or her intensity level in regard to the RPE scale is the same as yours. For example, if you tell your client to exercise at a level 10 but because the individual does not have a high tolerance for intense training (i.e., pain tolerance), the client stops the exercise at what appears to be a level 5 or 6, you would be better off using a set rep range. The *perceived intensity assessment (PIA)* can be used to help assess a client’s intensity threshold.

Time

Utilization of *time* versus *repetitions* to prescribe strength-training set durations is useful for any exercise, but especially for exercises that take a relatively long time, where counting reps would become monotonous. Using time is also helpful for isometric exercises such as a wall sit where there are no repetitions.

STRENGTH-TRAINING MACHINES vs. FREE WEIGHTS

Among fitness professionals and enthusiasts, there is a common debate over which strength-training equipment is better for individuals: machines or free weights.

Strength-Training Machine: characterized by an apparatus that moves in a set range of motion (e.g., chest press machine).

Free Weight: dumbbell

When discussing this comparison, we will include body weight and any exercise apparatus that does not move in a set ROM into the *free weight* category.

In terms of how different equipment in a gym setting is viewed, many individuals like to categorize equipment and/or exercises in a hierarchy based on an individual's level of proficiency. For example, most people identify the chest press machine for beginners when targeting the chest muscles whereas the bench press is best for intermediate exercisers and dumbbell/cable chest presses are best for advanced exercisers. As you can see, the progression is solely based around the level of stability that the exercise apparatus provides. Less stability equals more advanced.



Figure 9.2 Dumbbells

This certification denotes exercises that require machines sparingly because it is assumed that not all clients and coaches will have access to a fitness facility. However, the exclusion of strength training machines (also called *selectorized machines*) from this certification is not because they are not useful or are only for beginners. The selection of an exercise or piece of equipment should be based on many factors and not just the perceived experience level of a client. When choosing what if any exercise equipment to use, first assess what equipment is at your disposal, then assess what you are

trying to accomplish with your client.

Modifications

One thing to remember is that strength machines can also be “modified” to be more advanced. Let’s again discuss the chest press as an example. The standard way to do this exercise is to position the back against the back support and push out on the handles after selecting the appropriate weight. An advanced modification would be to sit forward on the seat so that the back is not being supported. This modification requires the core musculature to activate to maintain the upper-body posture while pushing out against resistance. A further modification would be to push with only one hand (weight must be lowered). This requires the core muscles to activate even more to resist torso rotation. Just because a client uses a strength machine does not mean it is only for beginners and cannot be progressed.

Selectorized strength equipment should be viewed from the standpoint of how the machines move rather than what muscles they state it is for. Looking at selectorized machines from this point of view will open up your options if training clients in a gym setting. For example, a back row machine could easily turn into a chest/core exercise. This would be accomplished by the following:

Instead of sitting on the machine and pulling the weight toward the body, an individual would kneel beside the machine facing the opposite direction. Once in this position, the individual would press forward with one handle while not rotating the upper body. This is just one example of how a wide array of options open up when machines are viewed based on movement patterns.

If you are working with a client in a health club during a busy time, being flexible in regard to the utilization of equipment is often a necessity as a particular piece of exercise equipment that you might have planned to use might be in use.

AREAS OF FOCUS

While most all areas of the body could benefit from strength training in regard to running performance, several areas should be focused on:

CALVES

As noted in the *Running Mechanics* module, the stiffness of the calf muscles (especially the gastrocnemius) and Achilles tendon increases the performance of the ankle spring mechanism that assists with propulsion. Therefore, exercises that focus on strengthening this area will help improve an individual's running economy. Assuming there are no contraindications, ballistic exercises such as plyometrics would be beneficial.

FEET

Like the ankle, the arch of the foot also acts as a spring mechanism to assist in the propulsion of a runner. The arch of the foot compresses upon foot landing and then acts as a spring to reform the arch. While one can do exercises to focus on the arch of the foot, walking around barefoot (indoors) is a great way to help strengthen it. Additionally, performing unilateral barefoot plyometrics can greatly strengthen the foot arch.

CORE (LPHC)

As noted in the *Muscular System* module, the LPHC acts to control the whole body in regard to movement and stabilization. Since running with proper form requires both movement and stabilization of the LPHC, training this area of the body is important for running form and performance.

GLUTEALS

The gluteals, specifically the gluteus maximus and medius, are important for runners. The gluteus maximus is a very large and powerful muscle that generates a lot of force and therefore is largely responsible for the forward propulsion of a runner, assuming correct form is used. The gluteus medius acts to prevent knee valgus (inward movement of the knee) and stabilizes the pelvis when standing on one leg during the gait cycle.

❖ RESISTANCE-TRAINING CATEGORIES

POWER DEVELOPMENT

Power is the rate at which work is done. The equation for power is: **$P = W/T$** (P = power, W = work, and T = time).

Therefore the faster one can perform a movement or exercise (i.e., work), the greater the power output will be. This is the essence of power training.

Many people confuse *strength* with *power*. The definitions below help to correlate these two things.

Strength: the amount of force a muscle or muscles can produce

Power: the rate at which one can use his or her strength

As such, the best way to think of power is strength training but with the element of time added to it. The next example illustrates the difference between power training and strength training, as it relates to work and time:

Scenario

It takes a client 1.5 seconds to bench press 130 pounds one time.

There are two ways to increase power:

1. Bench press more weight in 1.5 seconds
2. Bench press the same weight in less than 1.5 seconds

These two examples show how increasing weight and/or decreasing time for the same movement can increase power output. If you have ever watched power-lifting competitions, this is the epitome of power, hence the name. Power lifters are able to lift tremendous amounts of weight and are able to do so very quickly. While this, of course, is not the goal of a runner, the factors that make a power lifter successful are the same that make a runner more powerful: strength, speed, and proper mechanics.

Ironically, while it is widely accepted that short, cardiovascular interval-type workouts increase the performance of endurance athletes, performing short, intense resistance-training exercises/sessions is viewed by many as being non-sport specific.

Plyometrics

The most common type of power training is called *plyometrics* and refers to explosive-type movements. Most any movement can be turned into a plyometric exercise. For example, doing push-ups where the individual pushes off as hard as possible so that the hands leave the ground would be considered a plyometric

exercise. While these types of movements are fairly basic looking, from a physiological point of view, they are quite complex and therefore are considered advanced exercises.

The characteristics of a plyometric movement are a fast stretching phase (often termed *loading phase*) of a muscle followed by a rapid contraction of the same muscle (110). There is a direct relationship between the speed/distance of the loading phase and the resulting muscular contraction. An example of this would be jumping off a box and then jumping back up onto another box. Landing after jumping off the box is the loading phase and jumping back up onto the next box is the explosive phase.

The primary purpose of performing plyometric movements is to train the body to recruit as many muscle fibers at the same time within the muscles doing the work. The recruitment of large amounts of muscle fibers simultaneously increases the amount of force production over a short period compared with standard strength training. The goal of conventional strength training is to increase the amount of work that can be done without regard to time. Running requires both long-term strength and explosive power. Therefore both conventional strength training and plyometric training are applicable to running. In the most direct application, running is a series of many plyometric movements.

A 2013 study by Hudgins et al. found that standing long jumps (plyometric exercise) benefited not only sprinters but middle- and long-distance runners as well (693). Therefore increased power is important to long-distance runners, and as such, plyometric training should be a part of any distance-running program.

Another study by Skovgaard et al. examined whether repeated 30-second sprints and heavy resistance training performed in succession would increase performance in distance runners. The study found that this training protocol improved distance-running performance (1,500m), running economy, and dynamic muscle strength (694).

Because of the explosive nature of plyometrics, they should be done only by those who have a solid strength-training base and are able to perform a movement with proper form throughout the prescribed range of motion. When starting a plyometric program, you should limit your client to just one or two plyometric exercises per session and limit the range of motion, acceleration, and duration of the exercise. The range of motion, quantity, and intensity level of plyometric exercises per session can be increased as the client progresses.

Make sure that the client warms up sufficiently and moves through the range of motion that will be done during the plyometric exercises.

Like any exercise, plyometrics can be self-injurious if done incorrectly. Common causes of injury include:

1. Not sufficiently warmed up
2. Improper form/poor biomechanics
3. Undiagnosed or overlooked preexisting musculoskeletal issues
4. Progressed too quickly

Awareness of these reasons will significantly reduce the chance of injury. Perhaps the most important aspect to avoiding injury during jumping plyometric exercises is to avoid lumbar and thoracic spinal flexion, especially during the acceleration and deceleration. This will ensure that the lumbar and thoracic aspects of the spine are supported throughout the exercise.

Correlation to the Stretch-Shortening Cycle (SSC)

As noted previously, running is a plyometric exercise – meaning, running is a series of explosive movements. While excessive vertical movement during running (i.e., bounding) is inefficient, shuffling is not efficient either. Therefore there is an element of explosiveness that is required for a running-gait pattern to be efficient.

This relates directly to the *stretch-shortening cycle (SSC)* discussed in *Module 7*. To recap, the SSC is representative of an eccentric contraction of a muscle followed by a rapid concentric contraction of the same muscle. In regard to running, the SSC is most representative of the calves and Achilles tendon unit. Assuming the ankle joint has full mobility, the stiffer the calves and Achilles tendon unit are, the more efficient a runner will be because of a greater energy return.

SPORT REPLICATION

Sport-replication training relates to how closely an exercise replicates an actual sport movement. It is the goal of sport-replication training to have the highest correlation possible between exercise and sport-specific movement.

In order to replicate running movement patterns, the first place to look are the angles of the body during sport-specific movements. For example, the most common hamstring machine in a gym is the prone leg curl. This machine puts an individual in a facedown position in which only the lower leg moves – not the upper leg (femur) or any other body part. However, when running, the hamstrings are activated not only by knee flexion, but also hip extension. Therefore, would you be able to increase running performance just by strengthening the hamstrings using the prone leg curl? Perhaps. However, by performing exercises that mimic the exact motions of running, the crossover benefits will be substantially greater.

What Is Functional Training?

Functional training is a term commonly used in the fitness industry. By definition, functional training means that training is done for a specific purpose relating to the goal of a client. The term *functional training* is often and erroneously used to describe any multijoint, multiplanar exercise.

Since the definition of functional training is that any and all strength training must relate to the goal of the client, *all* training should be functional.

With the exception of some corrective exercises that focus on specific muscles for the purpose of correcting imbalances, the most important thing to remember when incorporating strength training into a runner's program is that **you are not training specific muscles – you are training movements.**

While this certification lists a large number of exercises, the amount possible is infinite. Therefore, when designing a resistance-training program for your client, do not reference exercises found solely in this certification but elsewhere as well.

CORRECTIVE

Corrective exercises are characterized by postural and structural exercises. These exercises have a strong emphasis on reducing muscle imbalances. Without addressing this area, strength gains in other areas of the body will likely be minimized. This would be like putting up walls of a house where the foundation is cracked. While you could build the house with a cracked

foundation, it would be a weakened structure and at some point it would break down.

These exercises specifically address a biomechanical, range of motion, muscle imbalance, or force production issue(s). While most exercises could be deemed a corrective exercise, the distinguishing characteristic of a corrective exercise is that one or more of the aforementioned issues are the main reason for the exercise selection.

STRENGTH

While all of the aforementioned categories of resistance training increase strength, this category relates to exercises that are not selected specifically for the purpose of increasing strength in relation to sport replication, corrective exercises, or power development.

A push-up would typically fit into this category. The purpose of a push-up is to strengthen the chest, shoulder, triceps, and core musculature. Depending on how it is done, the push-up could fall into the power or corrective categories. However, it is most often used to increase general strength.

General Conditioning

This aspect of strength training is used to describe a generic and overall full-body training program, often the focus of someone beginning a strength-training program. The characteristics are:

- Relatively slow progression
- Medium repetition range (10–15)
- Low to medium resistance on the muscles
- Medium speed of two or three seconds per repetition
 - No explosive movements

The purpose is to slowly adapt the muscles and connective tissue to increased resistance. This equates to the base phase of the cardiovascular training program in that it serves as the foundation for more specific training in the future.

Core Strength Training

As noted previously, the term *core* is quite general and can be more accurately defined as the *lumbo pelvic hip complex (LPHC)* – as the muscles in this region are what make up the core, not just the abdominals as typically referenced.

A 2009 study by Sato et al. found that core strength training improved subjects' 5K run performance (691).

❖ PROGRAMMING

The programming aspect of strength training is often where many coaches run into problems. The two main areas that coaches tend to have issues with are progression and integration.

PROGRESSION

Like cardiovascular training, strength training does not adhere to the 10 percent increase rule. For example, let us say your client started off being able to bench press 130 pounds. Using the 10-percent-per-week rule, eight weeks into the program your client would be able to bench press 253 pounds. As you can see, this is not feasible, correct, or sport specific. Strength training typically progresses at a much slower rate than cardiovascular training and does not typically increase every week or every training session.

Muscles adapt much faster to strength training than connective tissue does. Therefore, the progression of strength training must be very gradual in regard to time, weight, speed, and repetitions to avoid connective tissue injury.

If you are working with a client who has never done strength training or has not done it for a while (over one month), you must be very careful to start slowly and with a low weight. You also might introduce just a few exercises for several repetitions for the first strength-training session. Remember, this is not a personal training session where it has to take an hour and consist of all weight lifting.

You should increase the weight and repetitions (or time) in very small amounts and only after the client has shown ability to perform at the current level without notable muscle soreness in the following days.

Structure

As with the cardiovascular aspect of training, an individual will benefit from strength training to a greater extent with structure than with random strength training.

The following chart illustrates how the phases of a cardiovascular-training program can resemble that of a strength-training program in regard to progression.

Cardiovascular	Strength
Base	General Conditioning
Development	Strength and Corrective
Peak	Power and Sport Specific

Figure 9.3 Relating strength to cardiovascular periodization

While the progression from one phase to the next is not set in stone, the overriding principle is that in order for a client to gain strength and power safely, there must be some form of progression.

While you can have an emphasis on a particular type of resistance training, there should not be exclusively one type performed at any given time.

As muscles and connective tissue do not rapidly increase in strength from week to week, there will likely be several weeks where the same weight and repetitions are repeated. This is both normal and acceptable. It is much better to go this route than to progress a client too rapidly and risk injury.

From a safety standpoint, the progression of strength training is as follows:

1. **General Conditioning:** midrange repetition range (10–15 reps), two or three sets, light to medium weight
2. **Strength/Endurance:** low repetition-high weight and high repetition-low weight, respectively
3. **Power:** medium weight/medium repetition, fast speed

The strength-training recovery phases (weeks) should coincide with the cardiovascular program's recovery phases.

While this is the general progression, your client may or may not progress to all these areas. For example, your client may stay in the general conditioning phase throughout the training program or the individual might not progress to the power phase because of an injury. The degree of progression is completely based on the individual.

Strength-Training Taper

There is not much research regarding strength-training tapering for runners. As such, the following are rough guidelines:

- For 5K, 10K, and half marathon running events, lower-body strength training should be eliminated.
- For marathons, lower-body resistance training can be done during the first week of the taper but in limited amounts (based on a three- to four-week taper). After the first week of the taper, lower-body resistance training should be ceased.

The aforementioned guidelines are noted specifically in regard to lower-body exercises. Core and upper-body exercises can still be done during the taper. However, it should be done in limited quantities and intensity.

Individuals should not begin a strength-training program or introduce high intensity (HIIT) during a race taper period.

MUSCLE SPECIFICITY

It is common to associate fitness in one area with fitness in another. For example, even if your client is an elite-level runner who can run a four-minute mile, if the person has never done strength training the client will be quite sore the day after an initial strength-training session. Conversely, if an avid weight lifter who doesn't do cardiovascular exercise ran six miles at a high level of intensity, the person would be in a world of hurt the following days. This also

holds true for exercise selection. If your client has been doing just squats for the legs and you change to leg extensions on a machine, more than likely the person's quadriceps will be sore the next day. The specific use of muscles, weight, speed of contraction, and repetitions all factor into the specificity of muscle use.

INTEGRATION

Your ability to successfully integrate resistance training into a training program is necessary to have a well-rounded program and thus a well-rounded client. A critical factor when integrating strength training into a cardiovascular program is not to diminish the client's ability to perform at the desired levels in other areas of the program. You must design a resistance-training program so that a client's muscles are not too sore to perform optimally when running.

How your client responds to strength training is individually based. These are general recommendations that should be modified on an individual basis.

- Because of muscle specificity, a sore muscle in one range of motion might not be affected in another range of motion. For example, an individual's hamstrings might be tight from performing squats the day before, but when running, the hamstrings do not bother the client.
- Strength training does not need to take place in a single session, but rather can be spread out to accommodate a client's personal, work, and training schedule.

As noted in the study cited at the beginning of this module (Rønnestad et al.) (287), resistance training just one day a week was shown to maintain lean muscle mass and increase power output.

Argument for and against the Integration of Strength Training

Some research challenges the effectiveness and validity of integrating cardiovascular and resistance training. The main point of contention is a lack of muscle strength development when done in tandem with a program designed primarily around the cardiovascular aspect.

Effects of Strength Training and Running Order

A study by Doma and Deakin looked to determine if either of the following scenarios was preferable (656):

1. Strength-training session followed by running (six hours between training sessions)
2. Running followed by strength training (six hours between training sessions)

What Doma and Deakin found was that in respect to running performance the day after the aforementioned training scenarios, the group that performed strength training prior to running showed greater running impairment (i.e., decreased running economy) versus the group that performed running followed by strength training.

AGAINST INTEGRATION

This argument was first brought to light in 1980 by Robert Hickson and was termed *concurrent training*. Hickson's research on concurrent strength and endurance training found that there was an inhibition of strength development when both strength and endurance training were done concurrently (181).

Hickson's 10-week study placed subjects into three groups:

1. Endurance
2. Strength
3. Combination of the two (Strength and Endurance)

An important aspect of this study was that the strength-training component was done with as much weight as possible (80 percent of 1-RM), and the repetition range was low (five repetitions) for all exercises (squat, leg press, leg extension, leg curl) except calf raises, which was 20 repetitions.

The strength and endurance group realized significant increases in strength until week six-seven, at which time strength peaked, then decreased during the last two weeks.

A 1985 study by Dudley and Djamil (182) also found that an inverse relationship might exist between strength training and endurance. This study integrated high-intensity cycling intervals and isokinetic strength testing. The primary finding was that a reduction of strength development occurred during high-velocity strength training but not at low velocities.

In both the Dudley and Hickson studies, it was determined that a lack of strength gain occurred only in muscles that were used for both endurance- and strength-training modalities. Therefore, the researchers determined that concurrent training affects muscles on an acute and local level, rather than systemically (181, 182).

FOR INTEGRATION

A 1999 study by Paavolainen et al. found that when explosive strength training was integrated with endurance training, performance in a 5K race improved compared with those who did not integrate explosive strength training (282). It is theorized by Paavolainen that this increase in performance was due to increased neuromuscular characteristics, which equated to increased running economy, as the subject's VO_2 max levels did not increase. In terms of running, a decrease in leg muscle activation often equates to longer foot contact with the ground. In this study, it is additionally theorized that the neuromuscular adaptations from explosive strength training decreased ground contact time because of consistently high muscle activation (i.e., force production). The result was faster 5K running times.

These findings were further corroborated by Mikkola et al., who found that concurrent endurance and explosive training increased anaerobic capacity without decreasing aerobic capacity (283).

Timing

Perhaps the most relevant study in regard to integration of strength training was by Sporer and Wenger (2003). They looked at the effect of the duration of time between aerobic exercise and strength training (185). Sixteen male subjects were divided into two groups. One group performed high-intensity cycling intervals while the other group performed a steady-state submaximal cycle test. The two groups performed an incline leg press (75 percent of 1-RM) at four, eight, and 24 hours after the cycle tests. Each strength-training assessment was

done at least 72 hours after the previous assessment to ensure fatigue from the prior test was not a factor. Below are the results:

	Control	4 Hours	8 Hours	24 Hours
Incline Leg Press	48 Reps	36 Reps	44 Reps	49 Reps

Figure 9.4 Results of strength-integration study (Sporer and Wenger)

Subjects also performed the bench press post-aerobic assessments with no difference in repetitions. This confirms previous studies that showed the negative effects of concurrent training are limited to muscles used for both endurance and strength applications.

Interestingly, there was no difference in the effect on strength-training repetitions between the submaximal and high-intensity interval groups.

Conclusion

Many of the studies that found concurrent training detrimental for strength gains assessed subjects with a high-weight/low-repetition range. Alternately, studies such as Sale et al.(184), which did not find strength to be compromised during concurrent training, evaluated subjects using a higher repetition/set range (15–22 reps, six–eight sets).

Therefore, it could be deduced that higher-repetitions (>15) protocols would have greater strength increases as compared with low repetition when training concurrently.

Additionally, none of the aforementioned studies on concurrent training implemented strength-training progression. It could be theorized that decreases in strength, such as noted in Hickson's 1980 study, could be lessened or eliminated by establishing some form of progressive structure.

While this area of sports science warrants more research, there are some things to consider when designing a training program:

- Concurrent training may decrease strength gains as compared with strength training alone.
- Concurrent training does not appear to negatively affect endurance (aerobic training).

- Overtraining may be a factor as to why concurrent training results in decreased strength gains.
- Strength training within eight–10 hours after an aerobic-training session (utilizing the same muscles) will likely result in diminished effectiveness.
 - Because of the duration of most individuals' workdays, a strength-training session could be implemented on the same day as another workout, assuming one is done in the morning and the other in the evening with at least eight hours in between.
- Muscles used minimally during aerobic training can be strength trained the same day with minimal or no reduction in effectiveness.
- A progressive, structured strength-training program should be implemented to realize the highest strength gains using a concurrent training model.
- Strength training one or two times/week will maintain lean muscle mass and increase power output.

Keep in mind that the majority of your clients will be coming to you not with the goal of becoming an elite runner, but rather using running events as a way to stay motivated and fit. Because of this, strength training is an important fitness component and one that needs to be a part of the training program.

Even if your clients see diminished strength gains because of integration with their aerobic training, they are still realizing strength gains that otherwise would not occur with no strength training.

❖ SUMMARY

- Resistance training is an integral part of a running program.
- Performing resistance training only during the off-season violates the *use it or lose it* philosophy.
- There is no such thing as “almost correct form.” Resistance training form is either correct or incorrect.
- Resistance training of large muscle groups stimulates the neuroendocrine response. This response increases testosterone and hormones (i.e., growth hormone) that stimulate muscle growth/strength.
 - Exercises that elicit the highest neuroendocrine response are those that involve large muscles groups (i.e., squat, dead lift).
- The genesis of any strength-training program is the *overload principle*.
- A muscle that is a *prime mover* is the primary muscle responsible for movement. A *secondary mover* is a muscle that assists the primary mover.
- Many endurance athletes do not strength train for fear of developing substantial muscle mass and, thus, lowering their power-to-weight ratio. This fear is largely unwarranted and inaccurate.
- Resistance training can functionally strengthen the LPHC.
- The *strength-endurance continuum* pertains to the number of repetitions in relation to the desired effect (i.e., gaining muscular endurance).
- *Plyometrics* are explosive movements characterized by a fast stretch followed immediately by a fast contraction.
- Resistance training programs should have a progressive structure.
- It is advised to have at least eight–10 hours between aerobic- and strength-training sessions to maximize strength gains.
- Muscles used minimally during aerobic training can be strength trained the same day with minimal or no reduction in effectiveness.
- Resistance-training programs can be used for many things other than increasing muscle mass. Following are some of these things:
 - Increase power
 - Reduce muscle imbalances
 - Replicate sport-specific movements